Supplementary Information

Soil acidity governs cropland phosphorus availability under future climate

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Supplementary Text 1 Data compilation

We compiled a global dataset of climate-change effects on agricultural phosphorus (P) cycling by conducting a comprehensive literature research, in both elevated carbon dioxide (eCO_2) and warming scenarios. Specifically, we searched all peer-reviewed publications before February 2025 on the Web of Science, using the search string: TS = (((elevated CO_2) OR (CO_2 enrichment) OR FACE OR warming OR (elevated temperature)) AND (phosphorus OR (phosphorus availability) OR (phosphorus cycling)) AND (agriculture OR cropland OR farmland) NOT (meta-analysis OR review OR incubation OR pot)).

To ensure rigorous screening and data quality, individual studies were selected based on the following criteria: (1) Field experiments must be conducted under valid manipulation infrastructures (including free-air CO_2 enrichment [FACE], open-top chambers [OTC], chamber and infrared heaters) featuring comparisons between ambient and climate treatments; (2) Studies must have measured soil available P or other quantitative cropland P components, such as soil total P, extracellular phosphatase activities, microbial biomass P, or plant tissue P uptake; and (3) Publications must provide descriptive statistics, explicitly reporting mean values, sample sizes (n), and standard deviations (SD) for both ambient and treatment groups. For studies where SDs were not directly provided, they were mathematically back-calculated from reported standard errors (SE) *via* $SD = SE \times \sqrt{n}$ or approximated by multiplying the mean values by the pooled average coefficient of variation[1]. All raw data were digitally extracted from the tables or figures using GetData software v2.25. When publications reported multiple sampling events or soil depths, data were extracted independently to preserve the fine-scale temporal and vertical variability. Following a standard PRISMA workflow, we finally collected 303 field observations from 49 climate-change experiments (82 publications) for the meta-analysis (Supplementary Fig. 1).

Supplementary Text 2 Bayesian meta-analysis

To systematically quantify the responses of global agricultural P cycling under climate change, we implemented a hierarchical Bayesian meta-analysis, which can explicitly identify key sources of variability and uncertainty for small data[2]. Specifically, the natural logarithm of the ratio (ln*RR*) was adopted as the effect size and calculated as: $\ln RR = \ln\left(\frac{X_{treatment}}{X_{ambient}}\right)$, where $X_{treatment}$ and $X_{ambient}$ represent the mean values of the response variable under climate-change treatment (i.e., eCO_2 and/or warming) and ambient conditions, respectively. The corresponding variance of $\ln RR$ for each independent observation was calculated using the *escalc* function in the R package *metafor*[3]. Then, effect sizes were further modeled as: $\ln RR_{ij} \sim \mathcal{N}(\mu + u_i, \sigma^2)$, where μ represents the overall mean effect, u_i is the random effect associated with the *i*th study, and σ^2 is the residual variance. Finally, models were executed using Markov Chain Monte Carlo (MCMC) sampling archived within R package *brms*[4]. Posterior distributions were generated using four independent MCMC chains with 6,000 iterations per chain for robust posterior inference. Effect sizes and slopes were summarized using posterior medians and 90% credible intervals, which optimally balance statistical conservatism and inferential sensitivity, minimizing Type II errors commonly caused by the inherent sample-size constraints[2]. An effect was considered statistically significant when the 90% credible intervals did not overlap zero. Rosenthal's fail-safe numbers were calculated to assess the potential publication bias that is considered absent when the fail-safe number exceeds $5n + 10$[5]. While limited publication bias was found, the responses of some variables (e.g., phosphatase and root P uptake) should be interpreted with caution (Supplementary Table 1).

Supplementary Text 3 Multiple regression and relative importance analysis

To decipher the critical drivers regulating cropland P availability under climate change, we implemented a tripartite statistical framework. This multi-facet framework integrates Bayesian parametric inference, information-theoretic model selection, and machine learning to separate the relative importance of climate (including temperature and precipitation), soil (including pH, total P stocks, and iron-associated organic carbon [Fe-bound OC]), and experimental factors (including duration, CO_2 magnitude, and warming magnitude) to available P alterations (Supplementary Table 2), ensuring that the identified hierarchies of drivers are stable and immune to the assumptions of single approach.

First, we performed an information-theoretic model selection based on the Akaike's Information Criterion corrected for small sample sizes (AICc). Using the R package *glmulti*[6], an ordinary least squares meta-regression model was systematically decomposed into all possible submodel formulations. The optimal model space was delimited using a strict threshold of ΔAICc ≤ 2 relative to the best-fitting candidate. Predictors exhibiting a sum of Akaike weights ≥ 0.8 across the entire exhaustive were identified as the most important predictors.

Second, the structural configurations optimized *via* the AIC-based screening were translated into the fully parametric Bayesian meta-regression framework (Supplementary Text 2). By running the selected multi-predictor matrices through parallel MCMC sampling in the *brms* environment[4], we extracted the partial regression coefficients and their corresponding 90% credible intervals for all variables. This step complements the model selection by explicitly assigning a physical directionality (positive or negative linear controls) and formal uncertainty bounds to each driver under precision-weighted observation variances.

Third, to cross-validate the parametric findings and safeguard against undetected non-linear behaviors or complex high-order interaction cascades, we executed a random forest analysis tailored specifically for meta-analytic outcomes using the R package *MetaForest*[7]. Unlike conventional machine learning methods, this approach implements a precision-weighted bootstrap sampling that dynamically assigns higher sampling probabilities to independent observations with smaller intra-study measurement variances[7]. We trained a forest ensemble composed of 10,000 regression trees, and the algorithm automatically calculated the permutation-based variable importance by assessing the mean decrease in prediction accuracy when the values of a given predictor were randomly shuffled across the dataset.

Supplementary Text 4 Sensitivity analyses of the confounding water-regime effects

To ensure that the main results are not confounded by redox conditions, we performed sensitivity analyses across water regimes. Specifically, we conducted subgroup analyses of available P changes by stratifying our global dataset into distinct redox-related categories: cropland types (paddy *vs.* non-paddy) and irrigation management practices (rainfed, conventional irrigation, and water-saving irrigation). Furthermore, we employed AICc-based model selection to quantify whether the paramount influence of initial soil pH persists when explicitly controlling for water-regime effects. On the basis of existing factors including initial soil properties, climate factors, and experimental setup, we constructed three alternative mixed-effects models to predict available P changes: (1) a full model with water regime incorporated as a predictor, (2) a model using data statistically controlled for the water-regime influence, and (3) a model constructed using the residuals after removing water-regime-related variances. Comparing these models allows us to rigorously verify the independent effect of initial soil pH against broader variations in redox conditions.

Supplementary Text 5 Identification of P crisis hotspots

We developed a spatial overlay analysis to delimit global agricultural P crisis hotspots and to potentially translate our mechanistic findings into actionable management. The pixel-by-pixel hotspot identification (0.1° × 0.1°) was executed by extracting and intersecting three independent global spatial data layers: (1) Soil acidity layer: Given our evidence that long-term Fe-mediated P occlusion are strictly restricted to acidic soils (Fig. 2a and Supplementary Fig. 3), we extracted global soil pH maps from the SoilGrids[8]. Grids were filtered to retain only acidic soils with a topsoil pH < 6.5; (2) Crop PUE layer: To account for the demand and potential vulnerability of crop P uptake, we integrated global cropland distribution layers with localized crop-specific PUE datasets[9]. Grids characterized by low intrinsic crop PUE (the lowest 25th percentile, here represented by maize) were extracted; and (3) Soil microbial PUE layer: To incorporate the microbial feedback loop and intensive plant–microbial P competition under climate change, we also utilized the high-resolution global map of soil microbial PUE (Supplementary Fig. 4a; adapted from Gao et al.[10]). Grids were filtered to isolate regions with low microbial P acquisition capacity. By performing a strict spatial intersection across these two/three raster layers, we successfully framed the most vulnerable hotspots for cropland P availability under future climate (Fig. 2c and Supplementary Fig. 4b).

Supplementary Table 1 Rosenthal's fail-safe numbers used to examine the publication bias

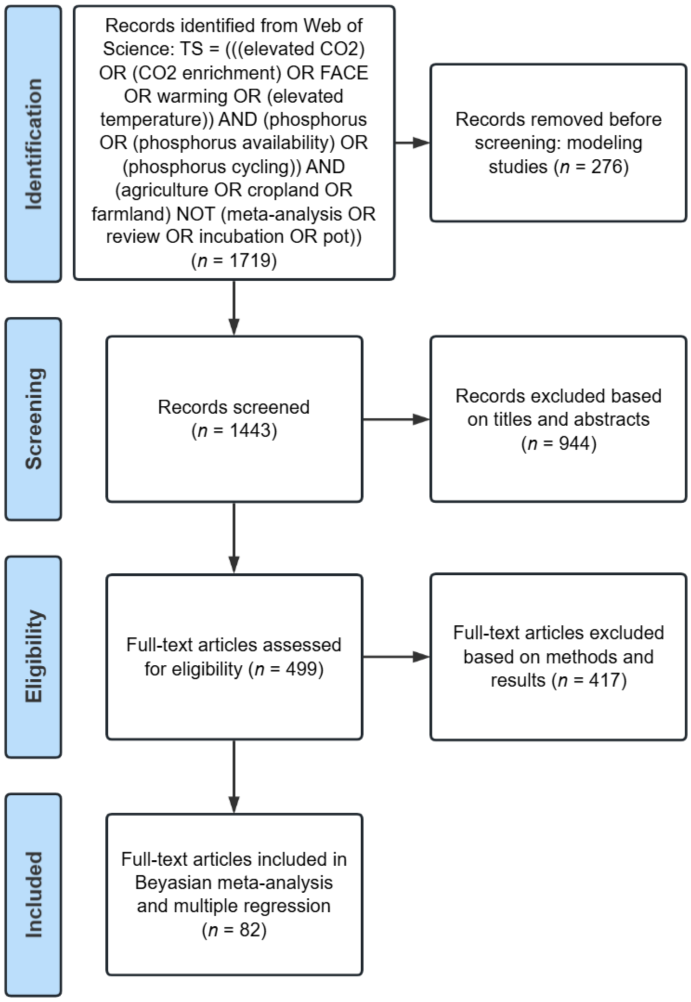
Variable	***n***	**Rosenthal's fail-safe number**	**5*n* + 10**	**Publication bias**
Available P	83	4661	425	no
Total P	44	1427	230	no
Phosphatase	17	0	95	yes
AMF	35	204	185	no
Shoot P uptake	142	33656	720	no
Root P uptake	27	90	145	yes
Shoot biomass	81	56260	415	no
Yield	55	25027	285	no

Note: There is no publication bias if the Rosenthal's fail-safe number is higher than $5n + 10$. n, the number of observations included in the Bayesian meta-analysis. AMF, arbuscular mycorrhizal fungi.

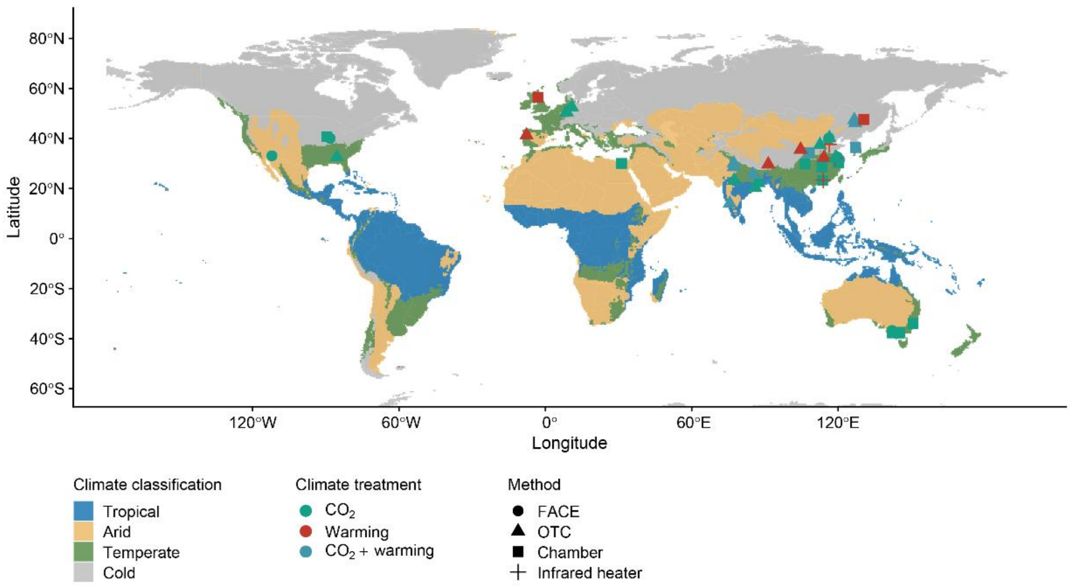
Supplementary Table 2 Predictors and their high-resolution maps used to fill with missing values and determine the available P changes in response to climate change

Predictor	**Unit**	**Original resolution**	**Data source**
MAT	°C	1 km	WorldClim v2.1[11]
MAP	mm	1 km	WorldClim v2.1[11]
pH	/	250 m	Soilgrids v0.2[7]
Total P stocks	kg m^{-2}	250 m	He et al. (2021)[12]
Fe-bound OC	g kg^{-1}	0.5°	Jia et al. (2024)[13]
Duration	year	/	Reported in publications
CO_2 magnitude	ppm	/	Reported in publications
Warming magnitude	°C	/	Reported in publications

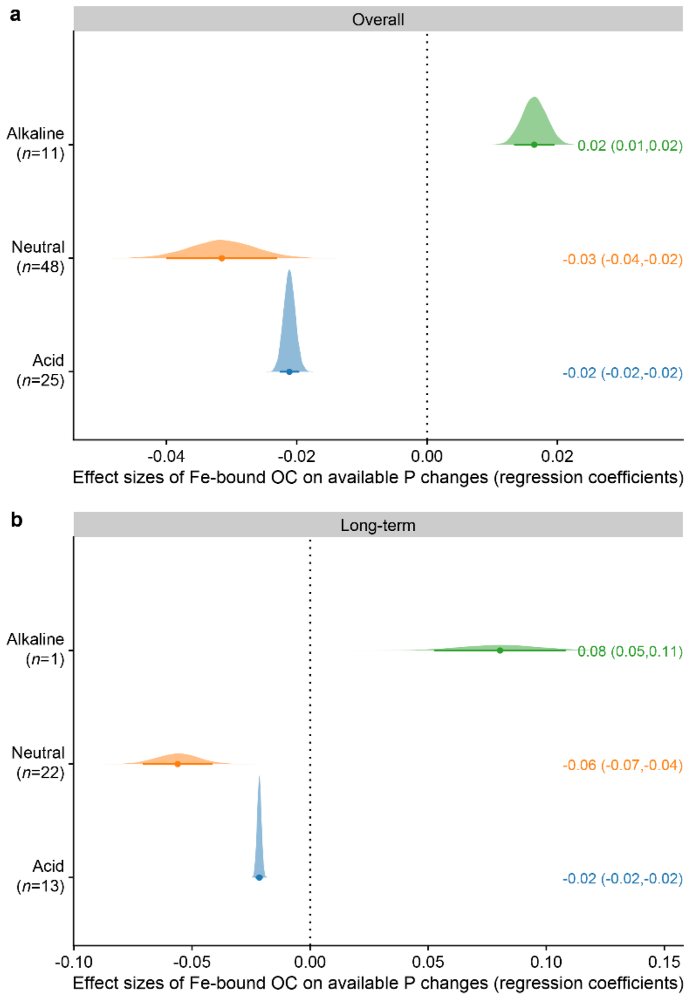
Note: MAT, mean annual temperature. MAP, mean annual precipitation. Fe-bound OC, iron-associated organic carbon. CO_2 magnitude and warming magnitude denote the increments of CO_2 and temperature compared to ambient levels, respectively.



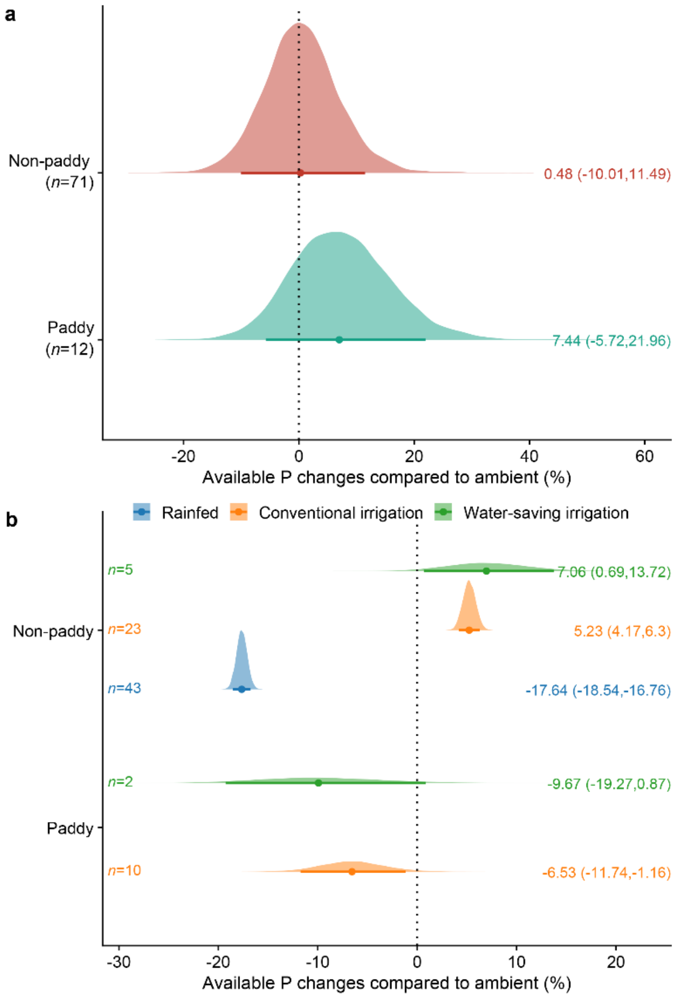
Supplementary Fig. 1 PRISMA diagram showing the workflow of literature screening and inclusion for the Bayesian meta-analysis and multiple regression.



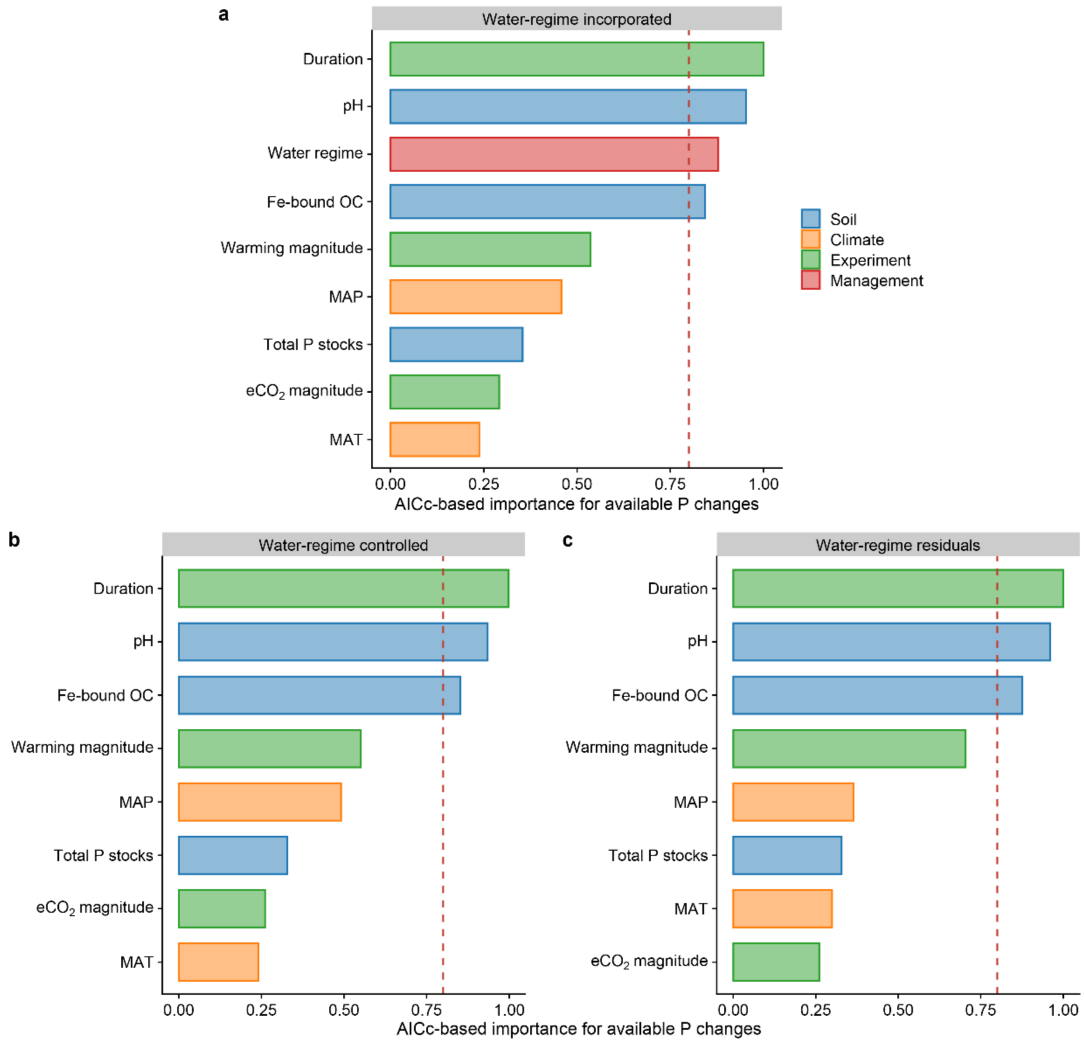
Supplementary Fig. 2 Global geographic distribution of 49 climate-change field experimental sites included in this study. Individual experiments are manipulated by multiple infrastructures, including FACE (free-air CO_2 enrichment), OTC (open-top chamber), chamber, and infrared heater, and are contextualized across tropical, arid, temperate, to continental regions.



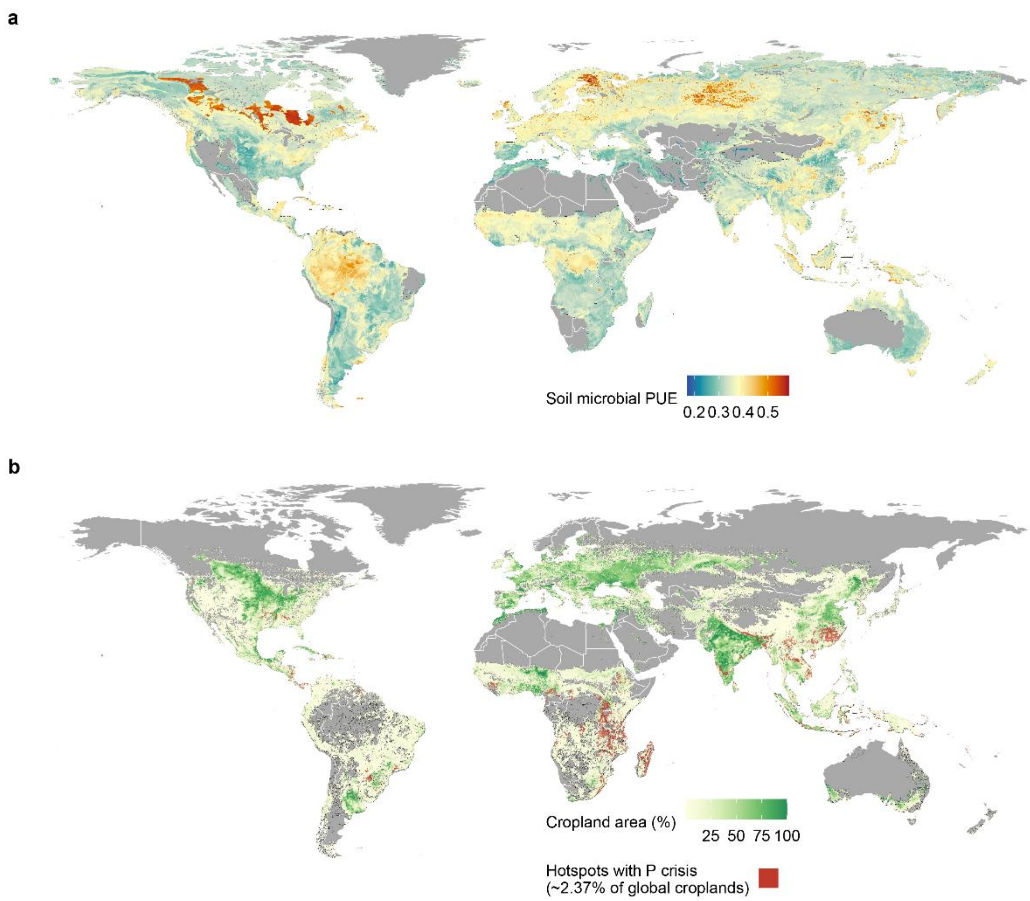
Supplementary Fig. 3 Effect sizes of Fe-bound OC on available P responses stratified across soil acidity categories in (a) the entire dataset and (b) long-term experiments (≥ 5 years). Shaded distributions represent the posterior density derived from the Bayesian meta-analysis, with central points and thick horizontal lines denoting medians and their 90% credible intervals, respectively. Numbers in parentheses indicate the sample sizes (*n*) for each category. Acid, pH < 6.5. Neutral, pH = 6.5–7.5. Alkaline, pH > 7.5.



Supplementary Fig. 4 Redox-dependent soil available phosphorus (P) responses to climate change. (a) Subgroup analysis of available P changes (%) under climate change scenarios across global paddy and non-paddy systems. (b) Influence of water management practices on available P changes (%) within paddy and non-paddy systems, categorized into rainfed, conventional irrigation, and water-saving irrigation conditions. Shaded distributions represent the posterior density derived from the Bayesian meta-analysis, with central points and thick horizontal lines denoting medians and their 90% credible intervals, respectively. Numbers in parentheses or on the left of the distributions indicate the sample sizes (*n*) for each system or each water management, respectively.



Supplementary Fig. 5 Sensitivity analysis evaluating the dominant role of soil pH after accounting for redox conditions. Corrected Akaike's Information Criterion (AICc)-based model selection assessing the relative importance of biotic and abiotic factors governing available P changes (n = 83) across three models: (a) full model with water regime explicitly incorporated as a predictor, (b) model using data statistically controlled for the water-regime influence, and (c) model constructed using the residuals after removing water-regime-related variances. Predictors include initial soil properties (pH, total P stocks, and iron-associated organic carbon [Fe-bound OC]), climate factors (mean annual temperature [MAT], mean annual precipitation [MAP]), and experimental setup (eCO_2 magnitude, warming magnitude, and duration), with/without water regime. Here, water regimes are classified into five combinations of land-use and water management (conventional irrigation, water-saving irrigation, or rainfed within paddy and non-paddy fields). The red lines in (a–c) denote the cutoff (0.8) to identify the most important factors.



Supplementary Fig. 6 Global patterns of soil microbial P use efficiency (PUE) and further delimited agricultural P crisis hotspots. (a) Global distribution of modeled microbial PUE adapted from Gao et al.[10]. (b) Hotspot identification for agricultural areas facing potential P crisis under future climate after incorporating plant–microbial P competition. Red grids delineate localized P crisis hotspots that are simultaneously characterized by acid soils, low crop PUE, and low microbial PUE, accounting for approximately 2.37% of the world's total cropland area.

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